

DAEN/ISEN 427 In class assignment 4

Due: Tue Oct 21 9:35 AM [10 pts]

Metropolis–Hastings Algorithm

Name/s: _____

Date: Thu Oct 16

Objective

This activity helps students build an intuitive understanding of the **Metropolis–Hastings (M-H) algorithm** by simulating it using a ball (football or baseball) passed among students around a table. Students will learn the key components of the algorithm:

- Generating proposals,
- Computing acceptance probabilities,
- Sampling from a target distribution.

Learning Outcomes

After this activity, students should be able to:

- Explain how the M-H algorithm generates samples.
- Interpret the acceptance probability as a balancing mechanism.
- Understand how sample frequencies approximate a target distribution.

Background

The M–H algorithm generates samples from a target distribution $P(x)$ by performing the following steps:

1. **Proposal:** From the current state x_t , propose a new state x' using a proposal distribution $Q(x'|x_t)$.
2. **Acceptance:** Accept x' with probability

$$\alpha = \min \left(1, \frac{P(x')Q(x_t|x')}{P(x_t)Q(x'|x_t)} \right),$$

otherwise stay at x_t .

Symmetric Proposal Assumption

If the proposal distribution is **symmetric** (passing left or right with equal probability), the formula simplifies to:

$$\alpha = \min \left(1, \frac{P(x')}{P(x_t)} \right).$$

Setup

Divide students into groups of 5 seated around a table. Each student is assigned to a target probability:

Student	Target Probability $P(x)$
A	0.10
B	0.20
C	0.40
D	0.15
E	0.15

Procedure

1. **Initialize:** The ball starts with one student (e.g., A). This represents the current state $x_t = A$.
2. **Propose a Move:** The ball holder randomly proposes to pass the ball to a neighbor (left or right with equal probability). This is the proposal $Q(x'|x_t)$.
3. **Compute Acceptance Probability:** (use the symmetric proposal simplification)

$$r = \frac{P(x')}{P(x_t)}$$

4. **Decide:** Generate a random number $u \sim U(0, 1)$. $\rightarrow$ e.g., in Excel: =RANDBETWEEN(0,1)
 - If $u < r$, **accept** and pass the ball.
 - Otherwise, **reject** and keep the ball.
5. **Record:** Each round, record who currently holds the ball.
6. **Repeat:** Perform 20–30 rounds of passes.

Data Recording Table

Round	Proposal	Accepted? (Y/N)	Current Holder
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
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22			
23			
24			
25			
26			
27			
⋮			

Analysis / Discussion

After completing the rounds, count how often each student held the ball.

Student	Observed Frequency	Target $P(x)$
A		0.10
B		0.20
C		0.40
D		0.15
E		0.15

Questions (Q1-Q8 due in-class Tue Oct 21 9:35 AM)

1. Briefly explain the main goal of the Metropolis–Hastings algorithm.
2. Describe how the acceptance probability is calculated in the algorithm.
3. What happened when the acceptance probability was high or low?
4. Plot a histogram of observed samples versus the target distribution.
5. How closely did your observed frequencies match the target probabilities?
6. How might asymmetric proposals (e.g., biased to the right) affect results?
7. What happens if the proposal distribution is poorly chosen?
8. What real-world problems could benefit from the M-H algorithm?

Other questions (for reflection only)

1. Change the target distribution and observe new sampling behavior.
2. Try asymmetric proposals where the ball is passed right 70% of the time.
3. Explain how the Metropolis–Hastings algorithm relates to Markov Chain Monte Carlo (MCMC) methods.
4. **Definition:** pseudocode is a simplified, informal way of describing the steps of an algorithm using plain language and basic programming structures. It helps outline the logic of a program without worrying about specific programming syntax.

Question: write a pseudocode to simulate the Metropolis–Hastings (M-H) algorithm as a classroom activity where a ball is passed among students sitting around a table. The pseudocode should include steps for:

- (i) generating a proposal for who to pass the ball to,
- (ii) computing the acceptance probability based on a target distribution, and
- (iii) deciding whether to accept or reject the proposed pass to simulate sampling from the target distribution.